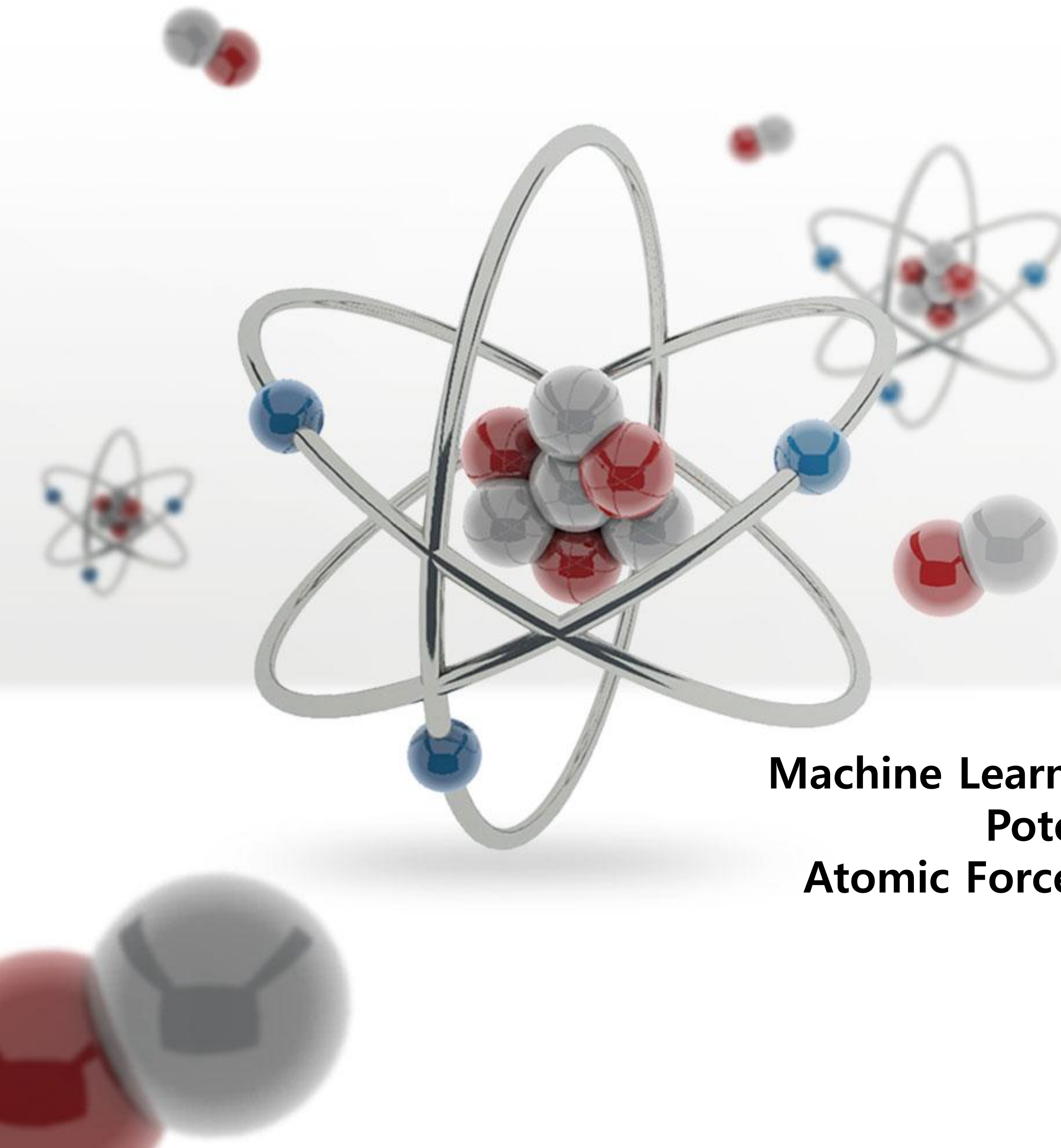
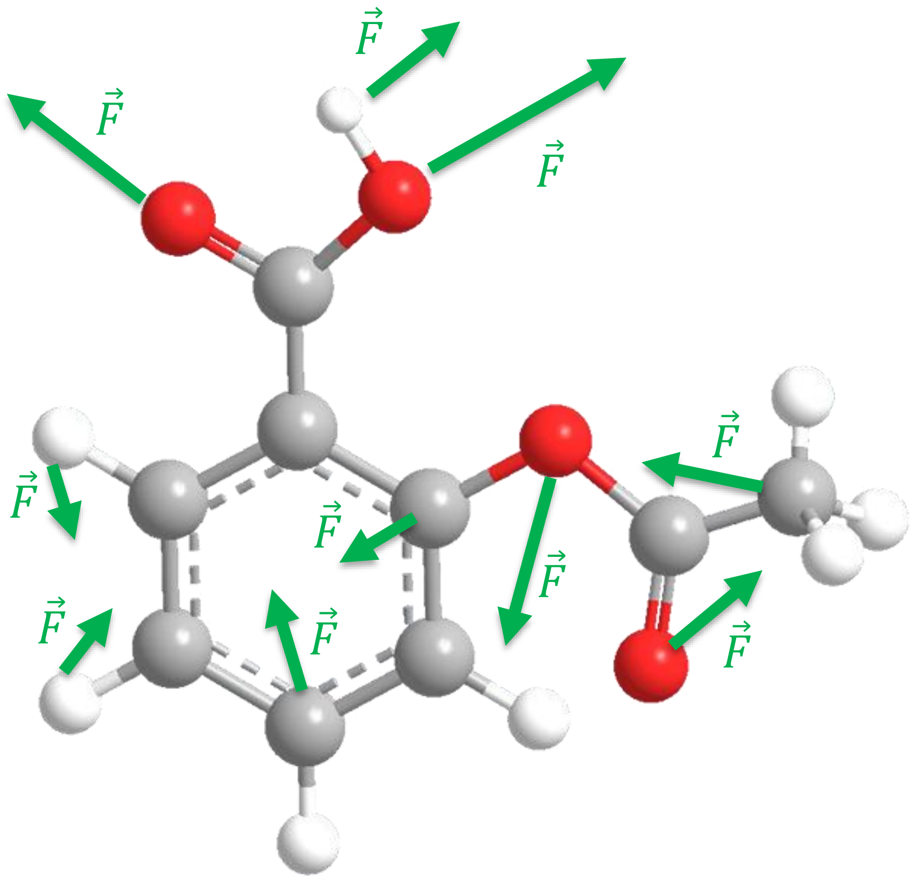


Machine Learning Force Fields: Predicting Potential Energy and Atomic Forces of Molecular Structures

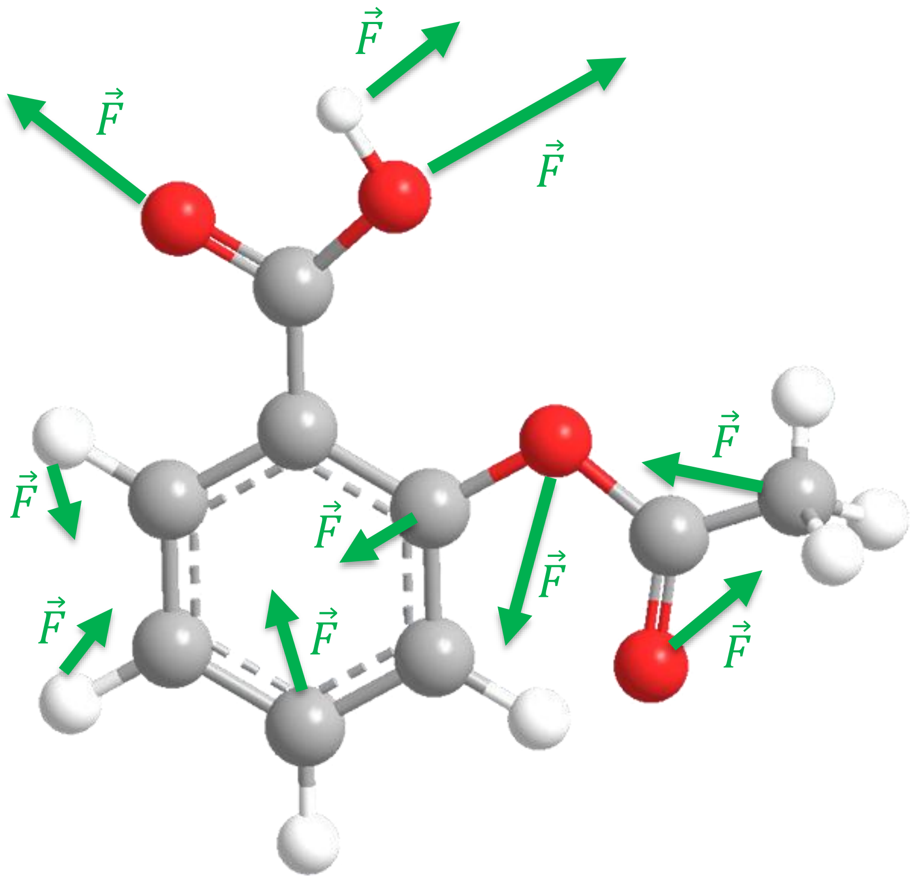
Amirarsalan Sanati

Problem Specification

Problem Specification



Problem Specification

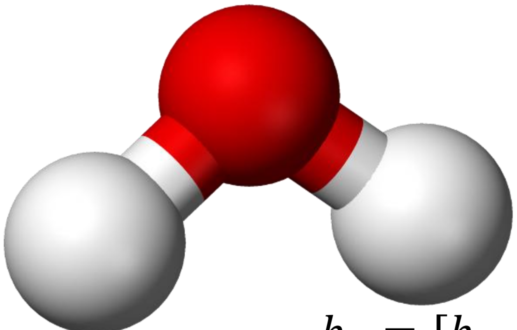


$E = \phi(Z, R) \longrightarrow \vec{F} = -\nabla_R E$

SchNet

SchNet

$h_1 = [h_{11}, h_{12}, h_{13} \dots h_{1n}]$



$h_2 = [h_{21}, h_{22}, h_{23} \dots h_{2n}]$

$h_3 = [h_{31}, h_{32}, h_{33} \dots h_{3n}]$

SchNet

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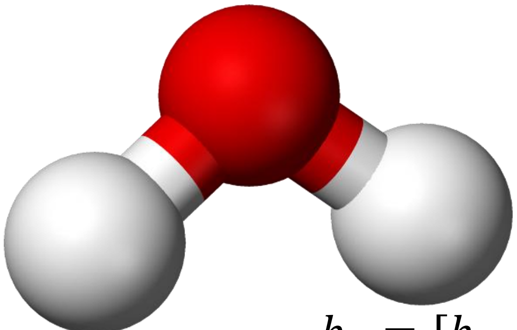

$h_2 = [h_{21}, h_{22}, h_{23} \dots h_{2n}]$

$h_3 = [h_{31}, h_{32}, h_{33} \dots h_{3n}]$

$$m_{ij} = \phi_{\text{msg}}(\mathbf{h_j}) \odot W(r_{ij}), \quad r_{ij} = \|\mathbf{r_i} - \mathbf{r_j}\|,$$

$$\mathbf{m_i} = \sum_{j \in \mathcal{N}(i)} m_{ij},$$

$$\mathbf{h_i^{new}} = \mathbf{h_i} + \phi_{\text{upd}}(\mathbf{m_i}),$$

SchNet

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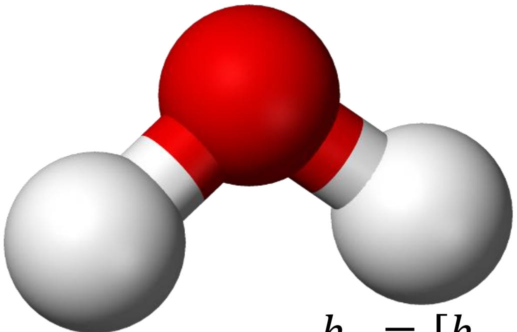

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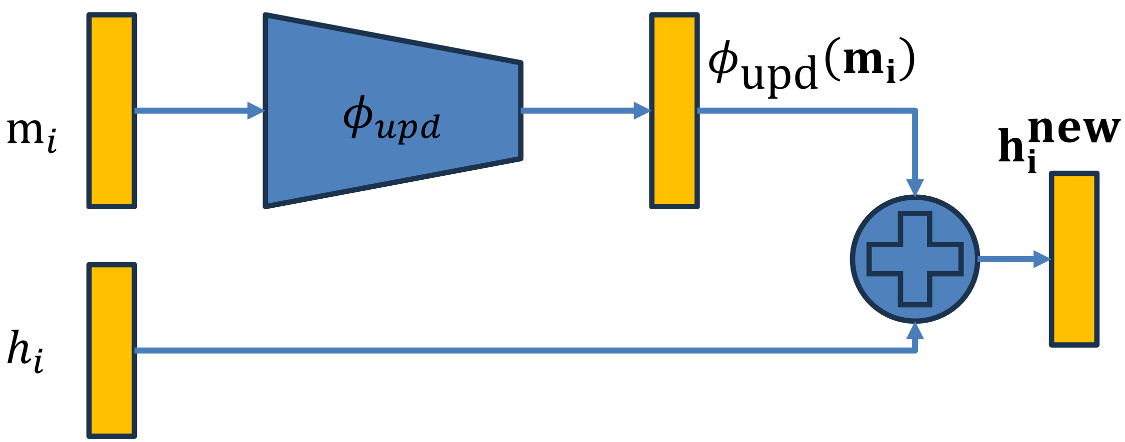

SchNet

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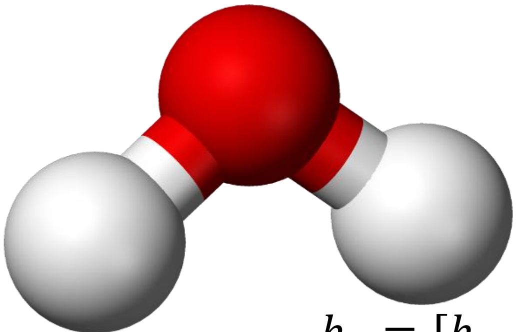

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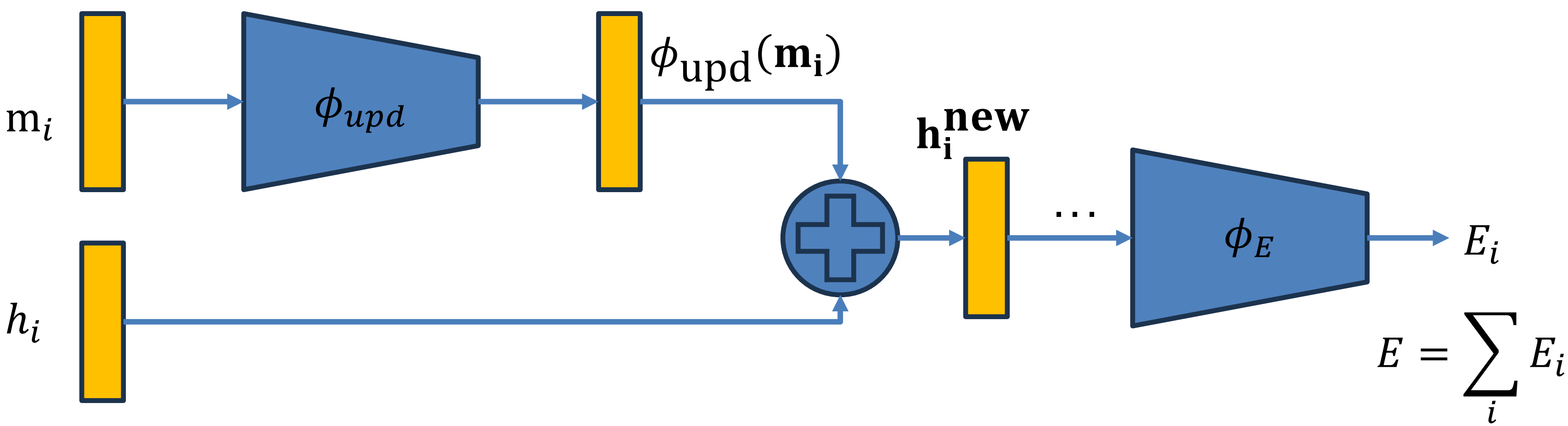

SchNet+

SchNet+

$$\mathbf{m_i^{sum}} = \sum_j m_{ij}$$

$$\mathbf{m_i^{avg}} = \frac{1}{|\mathcal{N}(i)|}\sum_j m_{ij}$$

$$\mathbf{m_i^{min}} = \min_j m_{ij}$$

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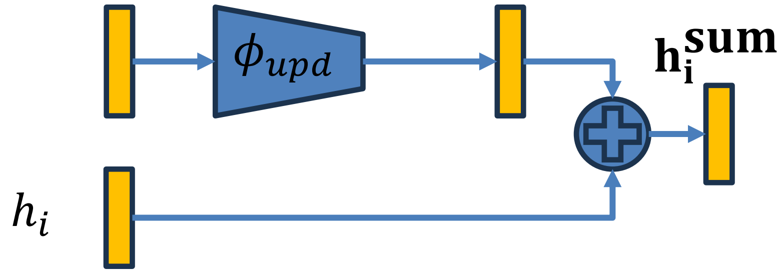
SchNet+

$$\mathbf{m_i^{sum}} = \sum_j m_{ij}$$



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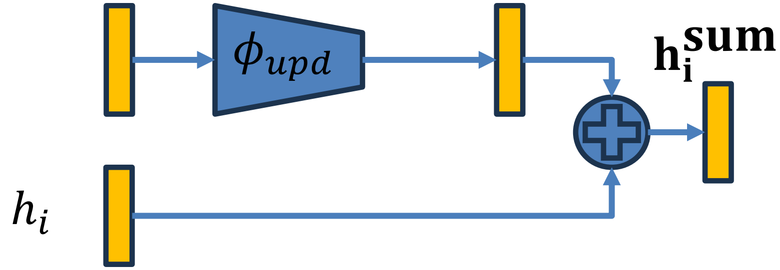
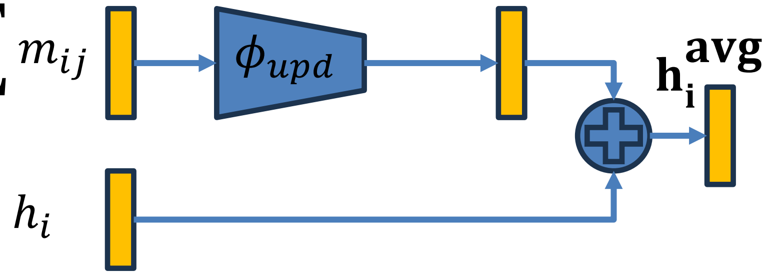


$$\mathbf{m_i^{avg}} = \frac{1}{|\mathcal{N}(i)|} \sum_j m_{ij}$$



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SchNet+

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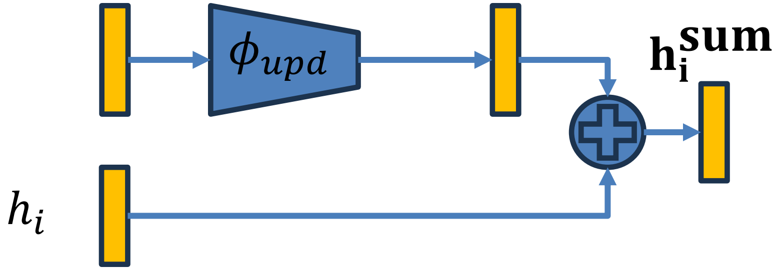

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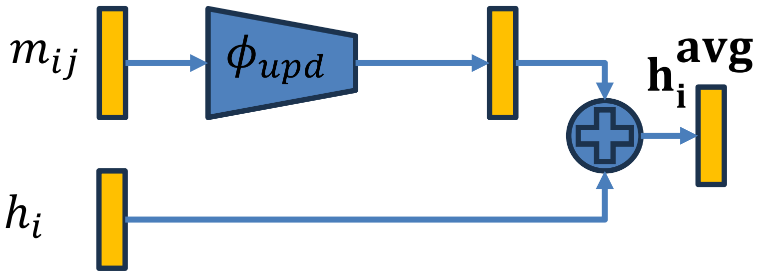
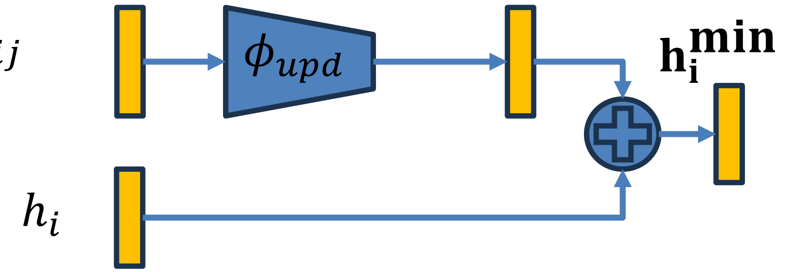


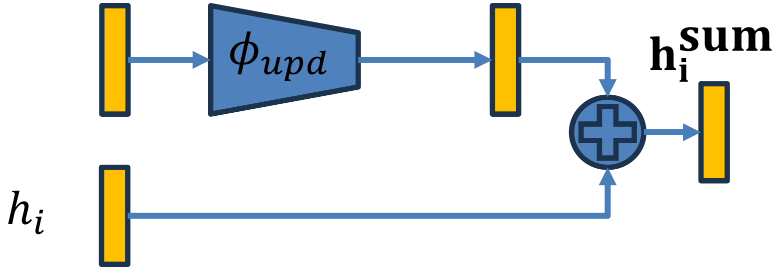
$$\mathbf{m}_\mathbf{i}^{\mathbf{min}} = \min_j m_{ij}$$



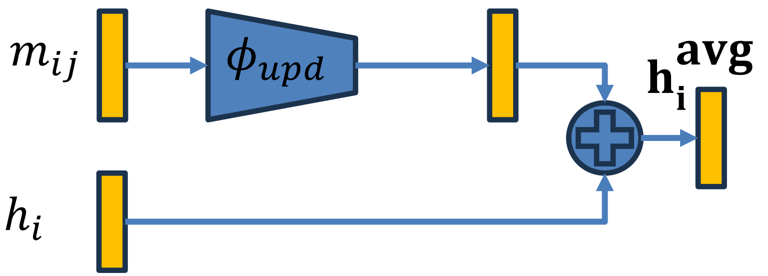
$$\mathbf{m}_\mathbf{i}^{\mathbf{max}} = \max_j m_{ij}$$

SchNet+

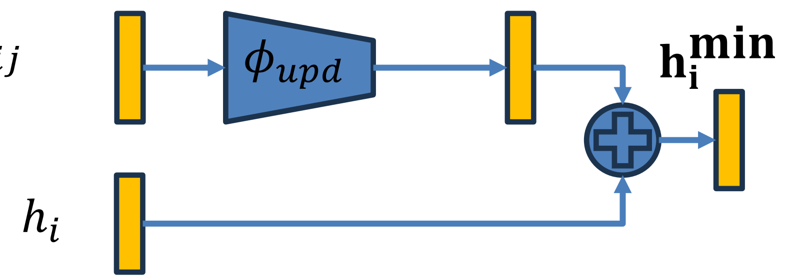
$$\mathbf{m}_{\mathbf{i}}^{\mathbf{sum}} = \sum_j m_{ij}$$



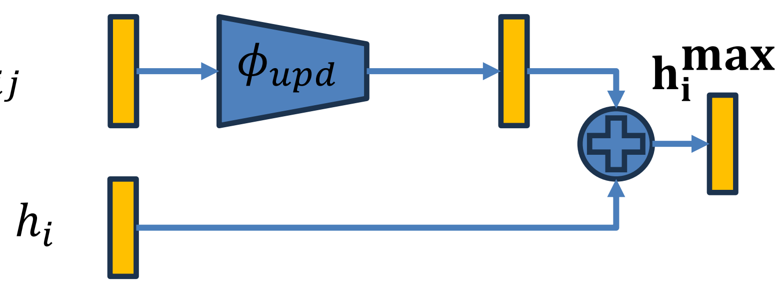
$$\mathbf{m}_{\mathbf{i}}^{\mathbf{avg}} = \frac{1}{|\mathcal{N}(i)|} \sum_j m_{ij}$$



$$\mathbf{m}_{\mathbf{i}}^{\mathbf{min}} = \min_j m_{ij}$$



$$\mathbf{m}_{\mathbf{i}}^{\mathbf{max}} = \max_j m_{ij}$$



SchNet+

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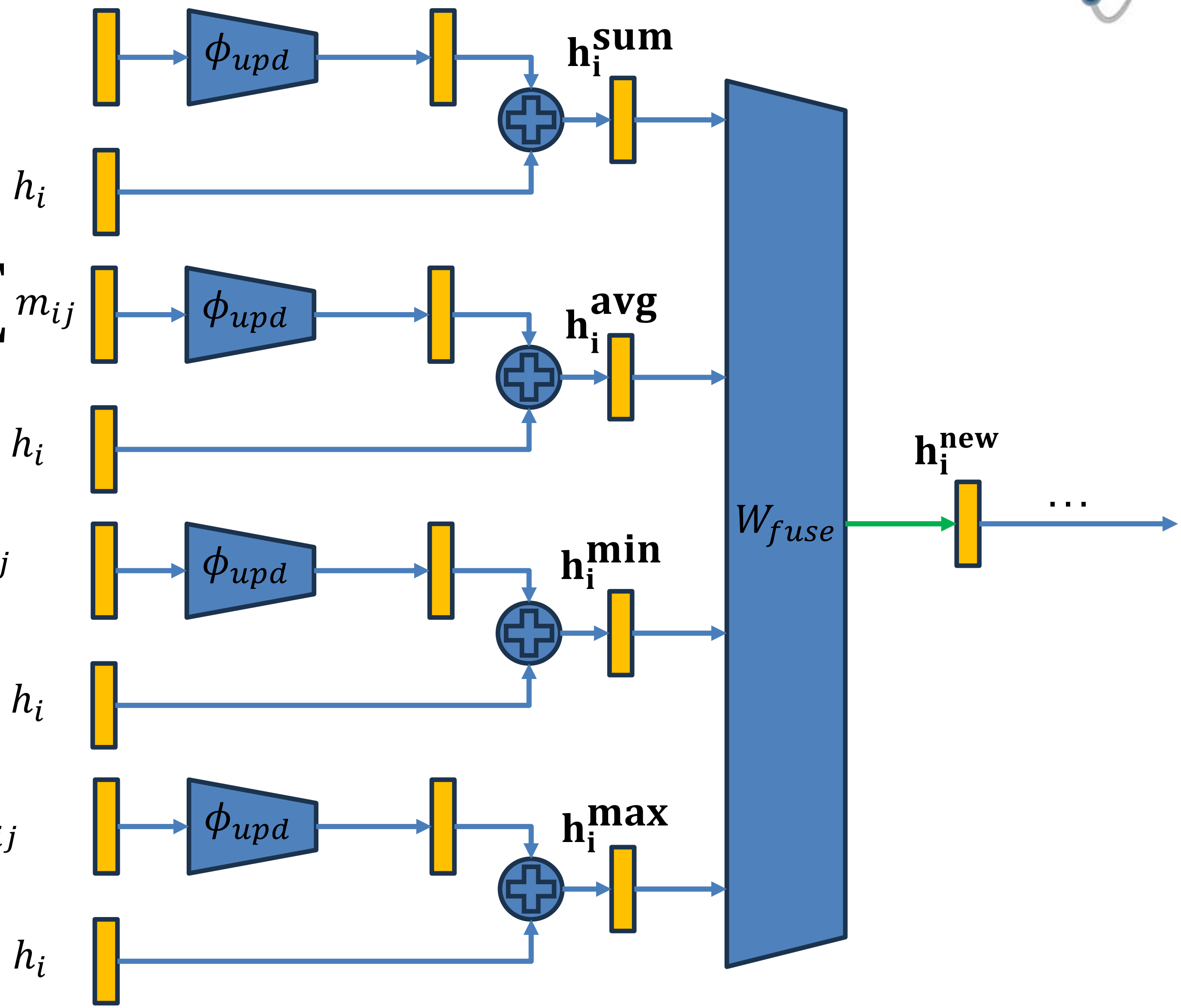


SchNet++

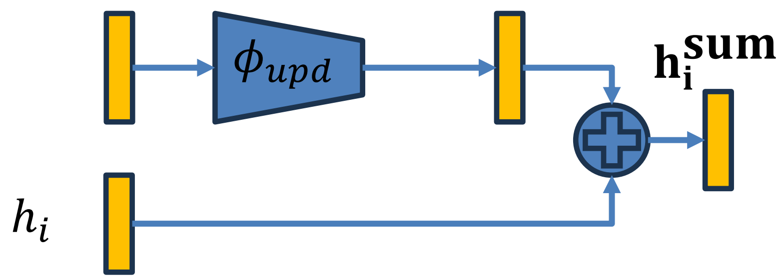
SchNet++

$$\mathbf{m_i^{sum}} = \sum_j m_{ij}$$

$$\left(\mathbf{m_i^{lse}}\right)_d = \frac{1}{\alpha_d} \log\left(\frac{1}{|\mathcal{N}(i)|}\sum_j \exp\left(\alpha_d (m_{ij})_d\right)\right)$$

SchNet++

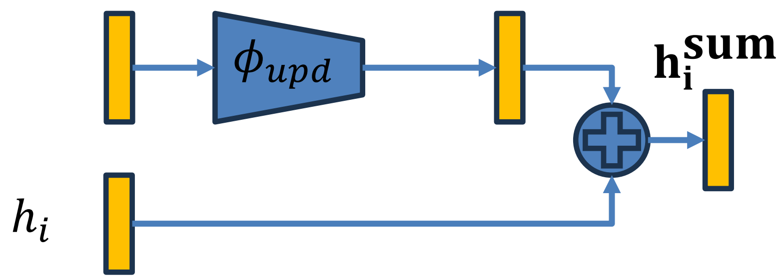
$$\mathbf{m}_{\mathbf{i}}^{\mathbf{sum}} = \sum_j m_{ij}$$



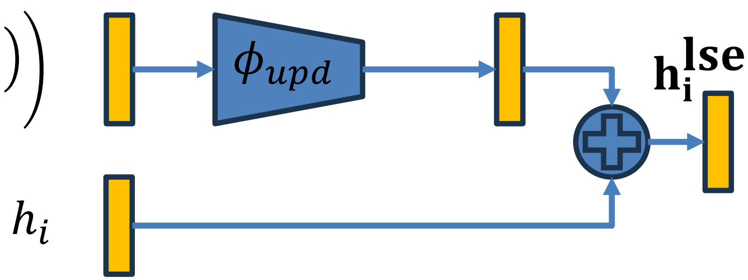
$$\left(\mathbf{m}_{\mathbf{i}}^{\mathbf{lse}}\right)_d = \frac{1}{\alpha_d} \log\left(\frac{1}{|\mathcal{N}(i)|} \sum_j \exp\left(\alpha_d \left(m_{ij}\right)_d\right)\right)$$

SchNet++

$$\mathbf{m}_\mathbf{i}^{\mathbf{sum}} = \sum_j m_{ij}$$



$$\left(\mathbf{m}_\mathbf{i}^{\mathbf{lse}}\right)_d = \frac{1}{\alpha_d}\log\left(\frac{1}{|\mathcal{N}(i)|}\sum_j \exp\left(\alpha_d\left(m_{ij}\right)_d\right)\right)$$



SchNet++

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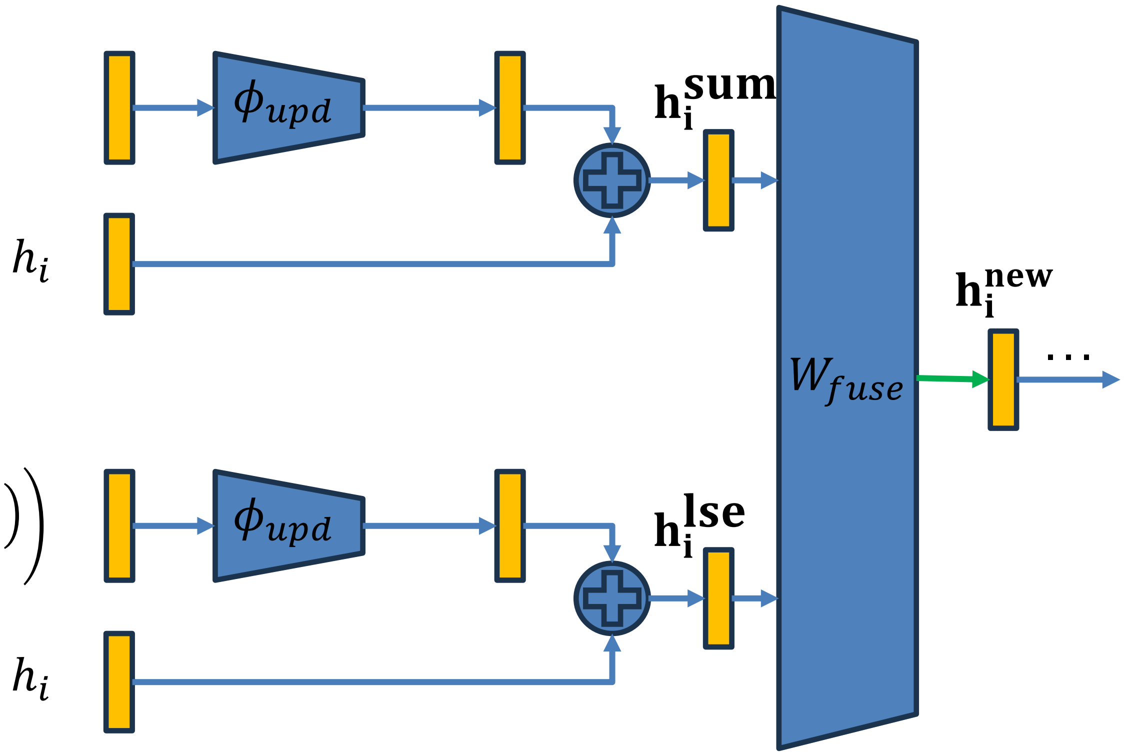
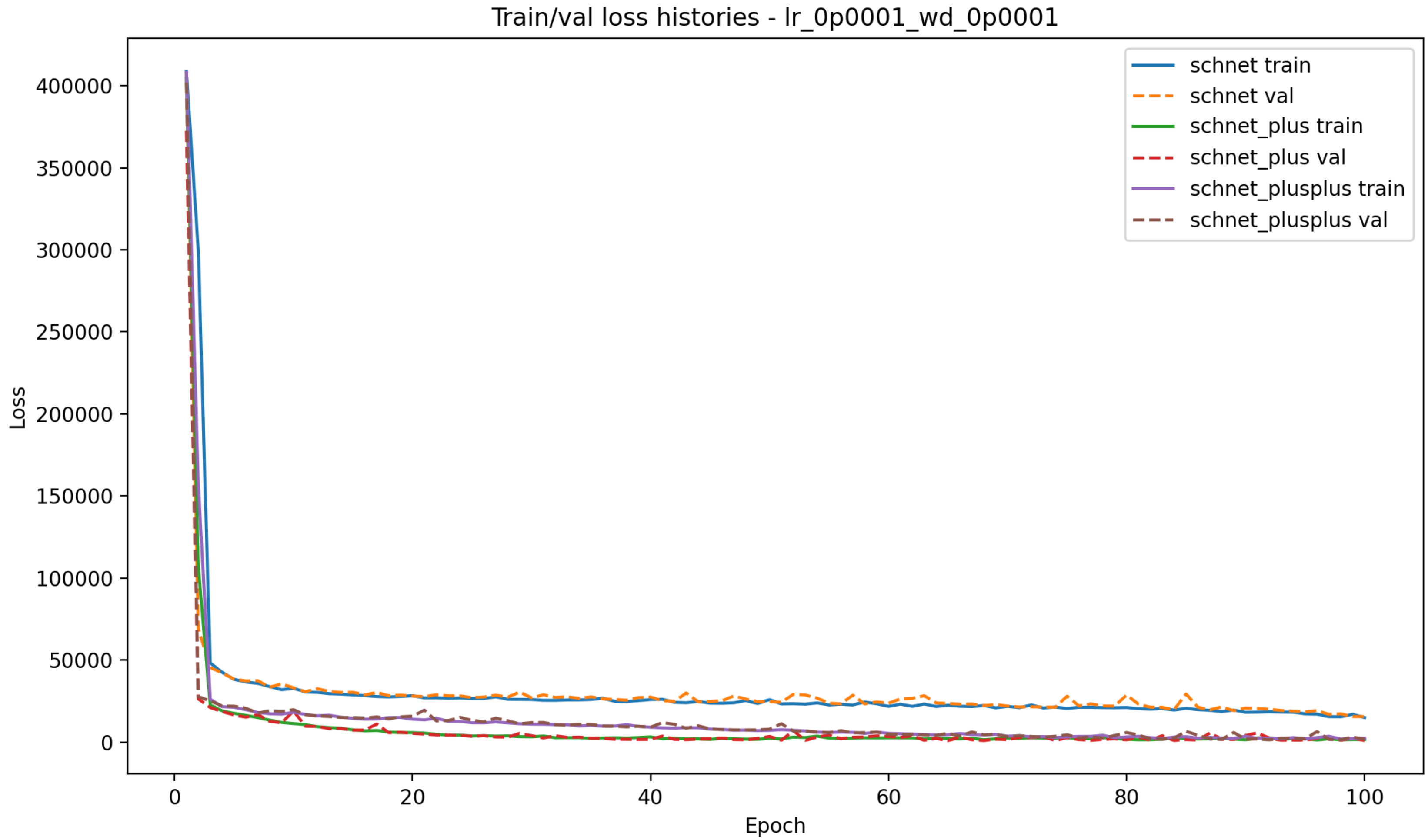
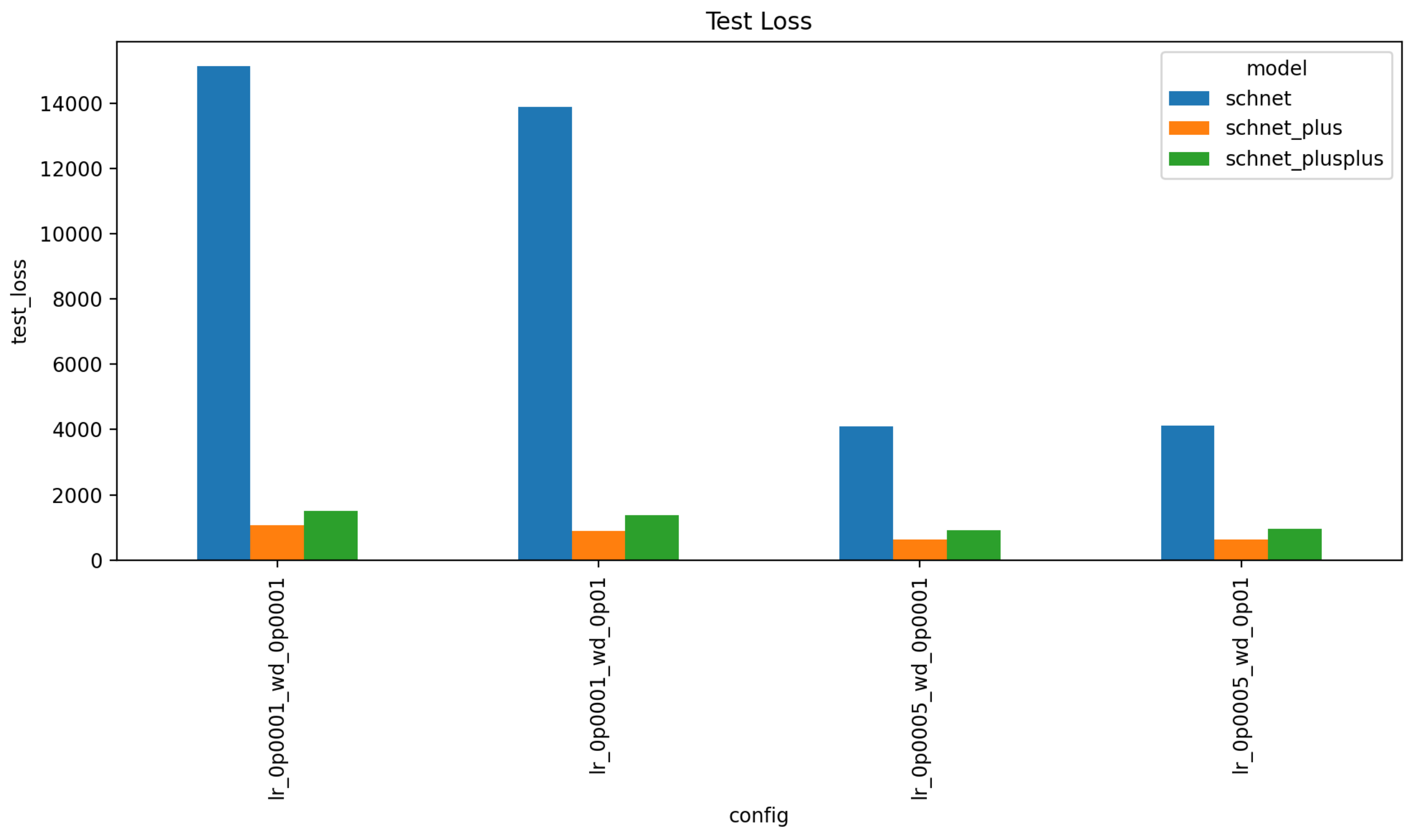


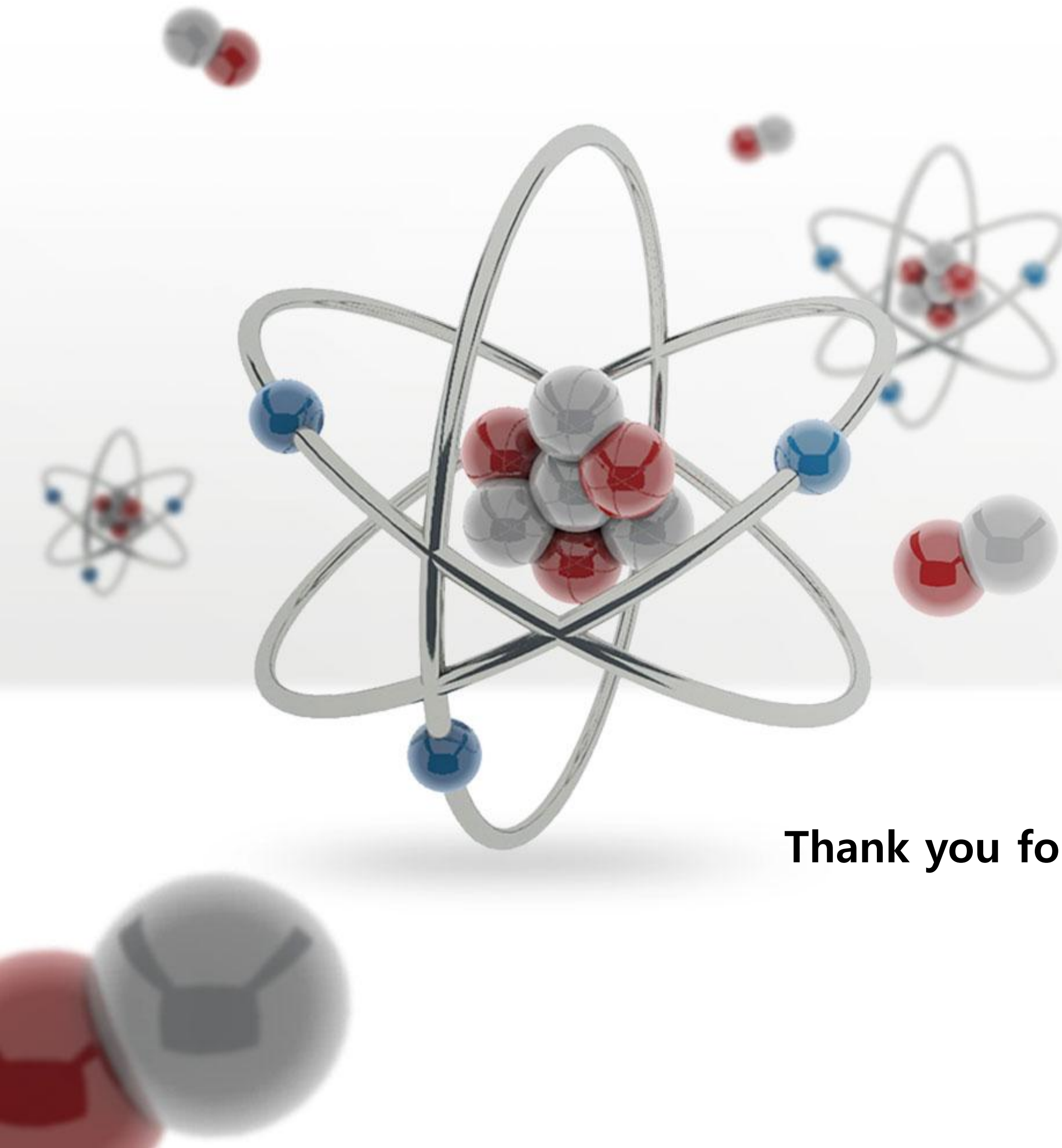
Train Results



Comparison



SchNet+	SchNet++	Metric
0.89	0.84	Relative Total Error Improvement
0.82	0.41	Relative Parameter Increase
1.09	2.05	Efficiency Score



Thank you for your time and attention